THE AI BRIDGE PROJECT | IIUI FMS

Website & Student Portal Implementation Plan

Updated System Architecture & Certification Workflow Blueprint

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1. Executive Summary & Revised Certification Criteria

The AI Bridge Web Application serves as the official digital management platform for the SDG Youth Campaign at International Islamic University Islamabad (IIUI). The platform streamlines event management, teacher training tracking, and student certification across three integrated user roles: Administrators, Ambassador Members, and Participating Students.

Key Workflow Update (Direct Download & 30-MCQ Evaluation):

- **Portal-Only Certificate Downloads:** Certificates will NOT be sent via email. To ensure security, data integrity, and verified user accounts, students must log into their personal portal dashboard to view and download their official PDF certificate.
- **3-Step Qualification Lifecycle:** To become eligible for certification, each student must complete three mandatory stages: (1) Pre-Survey, (2) 30-MCQ Assessment (selected dynamically from a 50-question course pool), and (3) Post-Survey.
- **Automated Progress Tracking:** The system dynamically tracks user progress across the Pre-Survey, MCQ Assessment score, and Post-Survey. The 'Download Certificate' button activates automatically on the student portal as soon as all three requirements are verified.

2. Student Certification Lifecycle (3-Step Assessment)

Stage	Requirement & Module Details	Verification & System Action
Step 1: Pre-Survey	Baseline knowledge and expectations survey completed prior to workshop/training.	Records baseline metrics in Supabase DB. Unlocks Stage 2 (MCQ Assessment).
Step 2: Course Assessment (30 MCQs)	Interactive online evaluation. System draws 30 randomized MCQs from a total course pool of 50 questions.	Evaluates core AI & NotebookLM comprehension. Passing score logged to user profile.
Step 3: Post-Survey & Direct Download	Post-training feedback & impact assessment. Unlocks direct PDF certificate download on portal.	Generates unique Certificate ID + QR Verification code. Certificate unlocked on User Portal for direct download.

3. System User Roles & Access Control

User Role	Target Group	Key Responsibilities & System Permissions

Admin	Project Lead & Core Tech Lead	Full administrative control over all events, question pool management (50 MCQs), student records, user role assignment, and system settings.
Member	IIUI SDG Ambassadors	Access to event check-in, attendance verification, pre/post survey monitoring, and student progress oversight.
User / Student	Trained Teachers & Students	Access to Student Portal (`/portal`), complete Pre-Survey, attempt 30-MCQ test, complete Post-Survey, and directly view/download PDF certificate.

4. Technical Architecture & Technology Stack

- **Frontend Framework:** Next.js (App Router, TypeScript) with React Server Components.
- **Styling & UI Components:** Tailwind CSS & shadcn/ui component library for accessible, modern UI.
- **Database & Authentication:** Supabase (PostgreSQL with Row Level Security, Supabase Auth with Google OAuth).
- **Assessment Engine:** Custom MCQ engine drawing 30 randomized questions from a 50-MCQ database table with score recording.
- **PDF Generation & Verification:** PDFKit / jsPDF engine generating branded certificates with unique verification QR codes linked to `/verify/[id]`.
- **Deployment & Hosting:** Vercel Platform with automated GitHub deployment pipeline.

5. Implementation Roadmap (Phases 1 to 5)

Phase 1: Foundation & Setup

Project scaffolding, Supabase database schema setup (Users, Events, MCQ_Pool, Surveys, Certificates), Next.js App Router structure, and Authentication integration.

Phase 2: 30-MCQ Engine & Survey Modules

Development of Pre-Survey form, 50-MCQ question bank setup, 30-MCQ dynamic quiz engine, scoring logic, and Post-Survey feedback module.

Phase 3: Student Portal & Portal Certificate Download

Student Portal (`/portal`) dashboard development displaying progress tracker (Pre-Survey -> MCQ Test -> Post-Survey) and one-click PDF Certificate download button.

Phase 4: Verification System & Public Route

Public QR Code verification route (`/verify/[id]`) allowing external employers or institutions to instantly verify certificate authenticity.

Phase 5: Testing, Security & Production Deployment

End-to-end testing across devices, Row Level Security (RLS) policies verification, Vercel production deployment, and project handover.

Project Lead Signature: _____ Date: _____

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